4.4 Wallis, Beta and the Factorial of a Half

WORKSHEET

The question

STEP II 1995, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s2-q4

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself What are u_0 and u_1? And roughly what does the graph of $\sin^n t$ look like on $[0, \frac{\pi}{2}]$ when n is large?

STAGE 1 A reduction formula

Ask yourself Integrate $\int \sin t \cdot \sin^{n-1} t \, \mathrm{d}t$ by parts, integrating $\sin t$ and differentiating $\sin^{n-1} t$. The new integrand contains $\cos^2 t$; how can you get back to powers of $\sin t$?

STAGE 2 An invariant

Ask yourself The relation says the quantity nu_nu_{n-1} does not change when n goes down by one. What is it equal to at $n = 1$?

STAGE 3 A squeeze

Pause and try Use $u_n < u_{n-1}$ and $nu_nu_{n-1} = \frac{1}{2}\pi$ to trap nu_n^2.

Try this next

1. **Wallis's product.** Using the explicit values of u_{2m} and u_{2m+1}, and $u_{2m+1} < u_{2m} < u_{2m-1}$, prove Wallis's product.

2. **A Beta integral.** Try STEP II 2013, Question 2 (step.maths.org/questions/13-s2-q2). It evaluates $\int_0^1 x^n(1-x)^n \, \mathrm{d}x$ with a reduction formula, then asks what the answer suggests when $n = \frac{1}{2}$: a first glimpse of the factorial of a half.

3. **The Beta function.** Try STEP III 2025, Question 1 (in OCR's download of the 2025 papers, www.ocr.org.uk/Images/750486-2025-past-papers.zip). It develops the general Beta integral $\int_0^1 x^{p-1}(1-x)^{q-1} \, \mathrm{d}x$, its symmetry and a recurrence, and uses them to evaluate an integral over $[0, \infty)$.

4. **Central binomial coefficients.** HARDERShow that $u_{2m} = \binom{2m}{m}\dfrac{\pi}{2^{2m+1}}$, and use $nu_n^2 \to \frac{\pi}{2}$ to deduce that $\binom{2m}{m} \approx \dfrac{4^m}{\sqrt{\pi m}}$ for large m. Check this for $m = 10$.

Further reading

- **Julian Havil, *Gamma*** (Princeton University Press). Includes the Gamma function and its history.
- **Philip J. Davis, "Leonhard Euler's integral: a historical profile of the Gamma function"**, *American Mathematical Monthly* 66 (1959). A classic, prize-winning article, readable with this section behind you.
- **Emil Artin, *The Gamma Function*** (Dover). A short, beautiful monograph for when you have done some analysis.

The full guided solution, and what the question leads to at university, is in Section 4.4:
https://beyondstep.keithmcnulty.org/ch04/wallis-beta.html